DAV SECTOR 4

Linear Programming Problems (LPP) — Interactive & Solution Suite

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Section 1: NCERT Standard Questions (Q1 to Q5)

5 Problems

Step-by-step mathematical derivation accompanied by real-time interactive plots.

Q1 NCERT Standard Maximization

Ex 12.1 Type

PROBLEM STATEMENT

Maximize $Z = 3x + 4y$ subject to constraints:

- $x + y \leq 4$
- $x \geq 0, y \geq 0$

Mathematical Solution Steps

Step 1: Convert inequality $x + y \leq 4$ to equation line $x + y = 4$. Intercepts are $(4, 0)$ and $(0, 4)$.

Step 2: Test origin $(0,0)$: $0 + 0 \leq 4$ (True). The feasible region is bounded in the first quadrant.

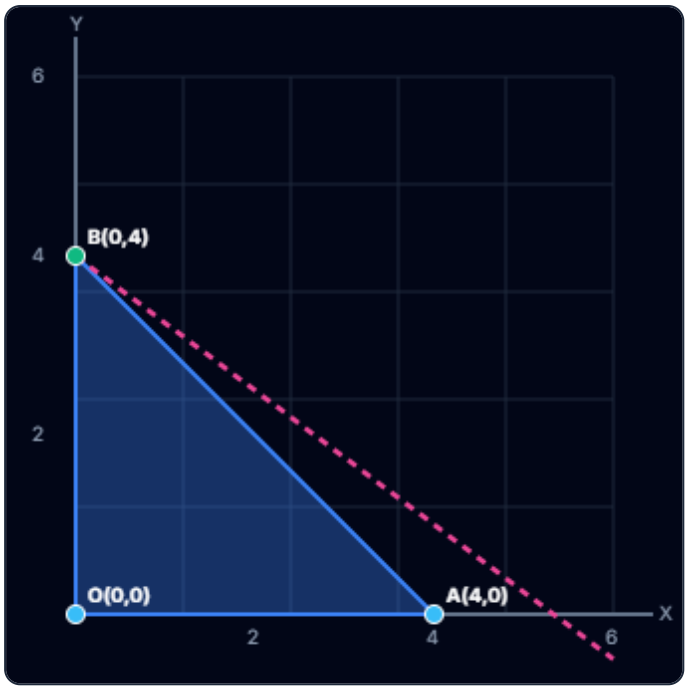
Step 3: Evaluate $Z = 3x + 4y$ at corner points:

Corner Point	Z = 3x + 4y Calculation	Value
O (0, 0)	3(0) + 4(0)	0
A (4, 0)	3(4) + 4(0)	12
B (0, 4)	**3(0) + 4(4)**	**16 (MAX)**

Conclusion: Maximum value of Z is **16** occurring at point **(0, 4)**.

INTERACTIVE REAL PLOT

Current Z = 16



Q2 **NCERT Standard Minimization** Ex 12.1 Core

PROBLEM STATEMENT

Minimize Z = -3x + 4y subject to constraints:

- x + 2y ≤ 8
- 3x + 2y ≤ 12
- x ≥ 0, y ≥ 0

Mathematical Solution Steps

Step 1: Plot boundary lines: $L_1: x + 2y = 8 \implies (8,0), (0,4)$ and $L_2: 3x + 2y = 12 \implies (4,0), (0,6)$.

Step 2: Solve simultaneous equations for point of intersection B:
$(3x + 2y) - (x + 2y) = 12 - 8 \implies 2x = 4 \implies x = 2, y = 3 \implies B(2, 3)$.

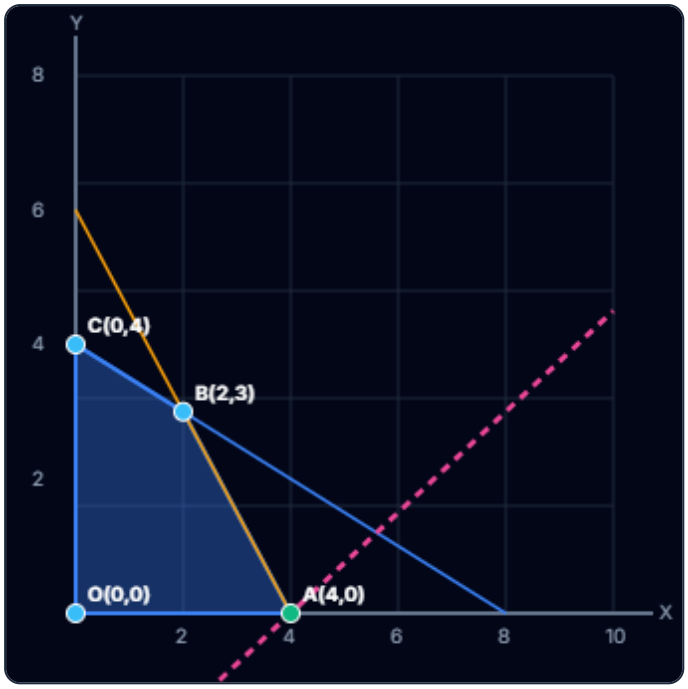
Step 3: Corner Point Evaluation for $Z = -3x + 4y$:

Corner Point	Z = -3x + 4y	Value
O (0, 0)	-3(0) + 4(0)	0
A (4, 0)	**-3(4) + 4(0)**	**-12 (MIN)**
B (2, 3)	-3(2) + 4(3)	6
C (0, 4)	-3(0) + 4(4)	16

Conclusion: Minimum value of Z is **-12** occurring at point **(4, 0)**.

INTERACTIVE REAL PLOT

Current Z = -12



Q3 **NCERT Intersecting Maximization** NCERT Pattern

PROBLEM STATEMENT

Maximize $Z = 5x + 3y$ subject to constraints:

- $3x + 5y \leq 15$
- $5x + 2y \leq 10$
- $x \geq 0, y \geq 0$

Mathematical Solution Steps

Step 1: Intercepts for $3x + 5y = 15 \implies (5,0), (0,3)$ and $5x + 2y = 10 \implies (2,0), (0,5)$.

Step 2: Calculate Intersection Point B by matrix/algebraic elimination:
Multiplying line 1 by 2: $6x + 10y = 30$; Line 2 by 5: $25x + 10y = 50$.
Subtracting: $19x = 20 \implies x = 20/19 \approx 1.05$. Substituting gives $y = 45/19 \approx 2.37$.

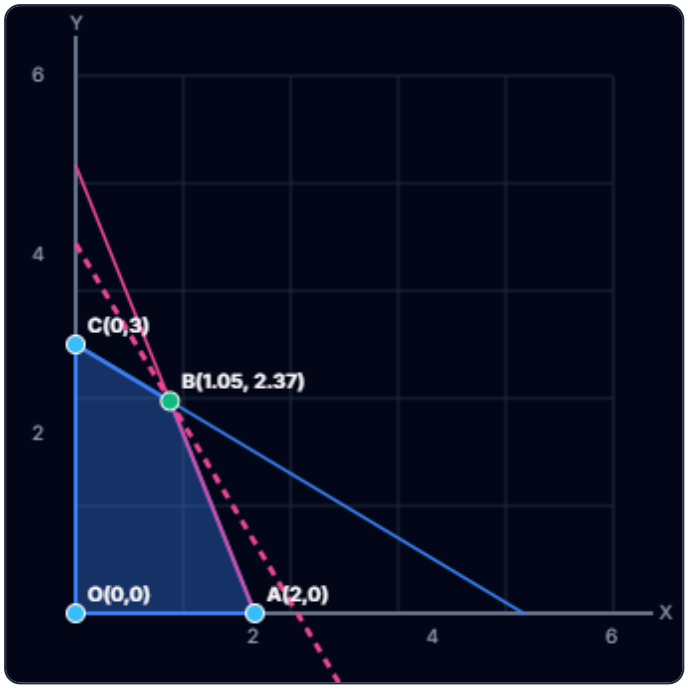
Step 3: Corner Point Table:

Corner Point	Z = 5x + 3y	Value
O (0, 0)	5(0) + 3(0)	0
A (2, 0)	5(2) + 3(0)	10
B (20/19, 45/19)	**5(20/19) + 3(45/19) = 235/19**	**12.37 (MAX)**
C (0, 3)	5(0) + 3(3)	9

Conclusion: Max Z is **235/19 (≈12.37)** occurring at **(20/19, 45/19)**.

INTERACTIVE REAL PLOT

Current Z = 12.37



Q4 Multi-Constraint Min & Max Problem

Ex 12.1 Advanced

PROBLEM STATEMENT

Minimize & Maximize $Z = x + 2y$ subject to:

- $x + 2y \geq 100$
- $2x - y \leq 0$
- $2x + y \leq 200$
- $x \geq 0, y \geq 0$

Mathematical Solution Steps

Step 1: Finding Corner Vertices by solving system of lines:

- A (0, 50): Intersection of x+2y=100 & y-axis
- B (0, 200): Intersection of 2x+y=200 & y-axis
- C (50, 100): Intersection of 2x-y=0 & 2x+y=200
- D (20, 40): Intersection of 2x-y=0 & x+2y=100

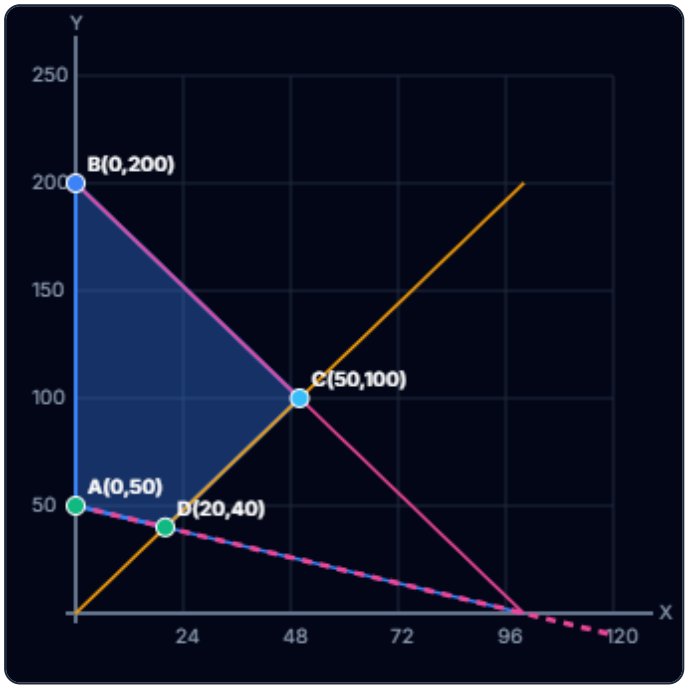
Corner Point	Z = x + 2y Calculation	Value
A (0, 50)	0 + 2(50)	100 (MIN)
B (0, 200)	0 + 2(200)	400 (MAX)

Corner Point	Z = x + 2y Calculation	Value
C (50, 100)	50 + 2(100)	250
D (20, 40)	20 + 2(40)	100 (MIN)

Conclusion: Min Z = **100** at two points (0,50) & (20,40) (multiple optimal solutions along line AD). Max Z = **400** at (0, 200).

INTERACTIVE REAL PLOT

Current Z = 100



Q5 Real-Life Manufacturing / Formulation Problem

Ex 12.2 Real-world

PROBLEM STATEMENT

A manufacturer produces two items A and B. Item A requires 1 hr of machine 1 and 3 hrs of machine 2. Item B requires 2 hrs of machine 1 and 1 hr of machine 2. Max hours available per day are 8 and 9 respectively. Profit on A is ₹250 and on B is ₹200.

Formulated LPP: Maximize $Z = 250x + 200y$ subject to:

- $x + 2y \leq 8$
- $3x + y \leq 9$
- $x \geq 0, y \geq 0$

Mathematical Solution Steps

Step 1: Intercepts for $x + 2y = 8 \implies (8,0), (0,4)$ and $3x + y = 9 \implies (3,0), (0,9)$.

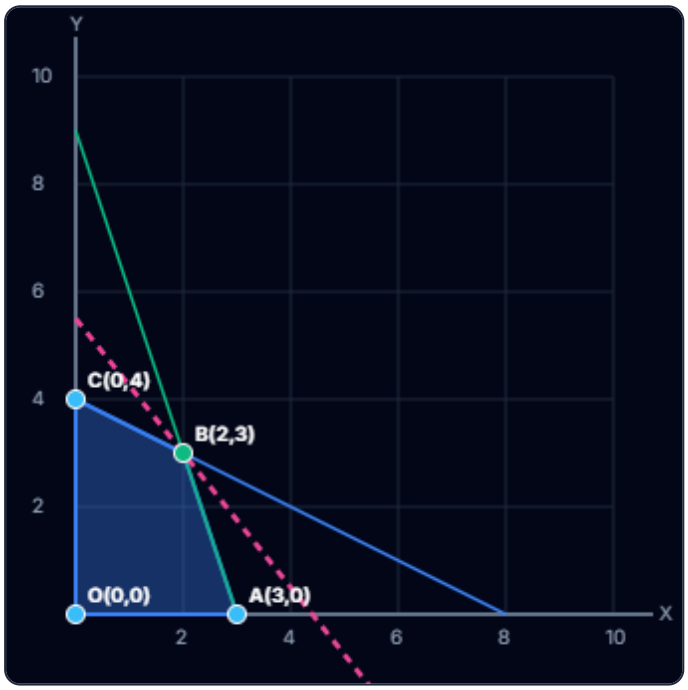
Step 2: Solve intersection B: $3(8-2y) + y = 9 \implies 24 - 5y = 9 \implies 5y = 15 \implies y = 3, x = 2$. Intersection is $B(2, 3)$.

Corner Point	Z = 250x + 200y	Value (₹)
O (0, 0)	250(0) + 200(0)	0
A (3, 0)	250(3) + 200(0)	750
B (2, 3)	**250(2) + 200(3)**	**₹1100 (MAX)**
C (0, 4)	250(0) + 200(4)	800

Conclusion: Max Profit is **₹1,100** by manufacturing **2 units of A** and **3 units of B**.

INTERACTIVE REAL PLOT

Current Z = 1100



Section 2: JEE Advanced Level Challenge (Q6)

High-difficulty non-trivial bounding analysis, exact area integration, & objective evaluation.

JEE Advanced Target

Q6 JEE Advanced Multi-Constraint Feasible Region & Area Analysis

Benchmark Problem

PROBLEM STATEMENT

Let feasible region R be bounded by inequalities:

- $2x + y \leq 8$
- $x + 2y \geq 4$
- $x - y \leq 2$
- $x \geq 0, y \geq 0$

Task: Determine the exact Maximum and Minimum values of $Z = 5x + 3y$ over region R, and compute the exact Area of Region R.

Rigorous Mathematical Solution

Step 1: Vertices of Feasible Quadrilateral ABCD:

- **A (0, 2):** Intersection of $x = 0$ and $x + 2y = 4$.
- **B (0, 8):** Intersection of $x = 0$ and $2x + y = 8$.
- **C (10/3, 4/3):** Solving $2x + y = 8$ and $x - y = 2$ simultaneously. Adding gives $3x = 10 \implies x = 10/3, y = 4/3$.
- **D (8/3, 2/3):** Solving $x + 2y = 4$ and $x - y = 2$ simultaneously. Subtracting gives $3y = 2 \implies y = 2/3, x = 8/3$.

Corner Point	Z = 5x + 3y Calculation	Value
A (0, 2)	5(0) + 3(2)	6 (MIN)
B (0, 8)	5(0) + 3(8)	24 (MAX)
C (10/3, 4/3)	5(10/3) + 3(4/3) = 62/3	20.67
D (8/3, 2/3)	5(8/3) + 3(2/3) = 46/3	15.33

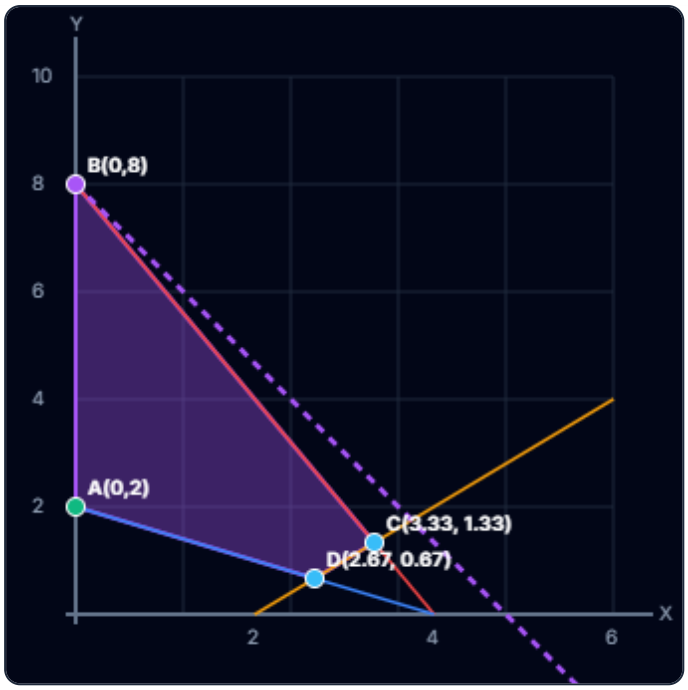
Step 2: Area of Quadrilateral Region R:

Splitting region along the vertical segment along the y-axis base AB (length = $8 - 2 = 6$ units): $\text{Area}(R) = \frac{1}{2} \times \text{Base} \times (x_C - x_D) = \frac{1}{2} \times 6 \times \left(\frac{10}{3} - \frac{8}{3}\right) = 3 \times \frac{2}{3} = \mathbf{5\text{ sq. units}}$.

JEE Final Result: Max Z = **24** at B(0,8) | Min Z = **6** at A(0,2) | Enclosed Area = **5 sq. units**.

DYNAMIC JEE PLOT

Current Z = 24.0


Y
10
8
6
4
2
B(0,8)
A(0,2)
C(3.33, 1.33)
D(2.67, 0.67)
2
4
6
X